

Supplementary Tables for Submission 991

CoLM 2026 Rebuttal: Full Supporting Tables

This document collects the full versions of tables summarized in our OpenReview responses. It supports those responses and is not required to follow them; each response is self-contained. All rebuttal-period numbers use fresh random seeds and can differ from the submitted paper by up to about 0.04 on coverage and width, within seed-to-seed variation. “Protocol A” denotes the submitted internal-split protocol (values quoted from the paper); “Protocol B” denotes the strict external 40/10/50 split used as a confirmatory robustness check.

Table A1. Fitted-model coverage and width under the strict external split (R1)

All three fitted-model methods (R2CCP, CQR, CHR) under Protocol A (paper) and Protocol B (strict external 40/10/50), 3 judges, 10 seeds per Protocol-B cell. Protocol-B values are mean $\pm$ standard deviation; Δ columns are paired t -test p -values on matched rebuttal-period runs. Coverage holds at the 0.90 target for R2CCP under both protocols; CQR over-covers throughout (weak quantile-regression features); CHR recovers nominal coverage under Protocol B once the finite-sample $(1 + 1/n)(1 - \alpha)$ correction is included (the standard split-CHR procedure of Sesia & Romano, 2021).

Method	Judge	Protocol A (paper)		Protocol B (40/10/50)		Paired p	
		cov	w	cov	w	Δ cov	Δw
R2CCP	LLaVA-Critic	0.900	3.05	0.894 ± 0.015	3.041 ± 0.095	0.772	0.093
R2CCP	Phi-4	0.891	3.13	0.890 ± 0.012	3.145 ± 0.076	0.377	0.908
R2CCP	Gemini	0.898	2.85	0.897 ± 0.012	2.887 ± 0.067	0.566	0.015
CQR	LLaVA-Critic	0.996	3.95	0.970 ± 0.011	3.60	0.002	0.002
CQR	Phi-4	0.997	3.98	0.993 ± 0.009	3.91	0.641	0.469
CQR	Gemini	0.996	3.95	0.977 ± 0.012	3.65	0.002	0.002
CHR	LLaVA-Critic	0.880	2.97	0.900 ± 0.014	3.29	0.002	0.002
CHR	Phi-4	0.896	3.16	0.902 ± 0.015	3.33	0.004	0.002
CHR	Gemini	0.880	2.80	0.906 ± 0.012	3.20	0.002	0.002

Protocol-A figures are quoted from paper Table 1 (cross-judge R2CCP and CHR rows), Table 2 (LLaVA-Critic CQR, 0.996/3.95), and Table 9 (cross-judge CQR, 0.996/0.997/0.996); rebuttal-period Protocol-A runs reproduce them within seed variation (CHR raw coverage 0.857 to 0.882, similarly below 0.90).

Table A2. Width-decile reliability on MLLM-as-a-Judge (R2)

Per-instance triples pooled across 3 judges and 10 seeds ($n = 83,894$ after dropping unparseable outputs), binned into deciles by raw CP width. Accuracy falls near-monotonically as width grows (one small inversion at deciles 5 to 7): the tightest decile reaches 43.9% exact accuracy and the widest 25.5%, a $1.7\times$ separation purely from the CP-width ordering.

Width decile (raw)	mean width	exact acc.	± 1 acc.	judge MAE
(0.35, 2.30]	2.12	43.9%	81.9%	0.84
(2.30, 2.53]	2.42	37.9%	77.1%	0.97
(2.53, 2.71]	2.62	35.1%	75.3%	1.01
(2.71, 2.90]	2.80	33.5%	76.3%	1.01
(2.90, 3.05]	2.99	31.9%	74.2%	1.06
(3.05, 3.14]	3.09	32.9%	75.2%	1.03
(3.14, 3.25]	3.20	31.9%	73.9%	1.05
(3.25, 3.41]	3.33	30.2%	71.5%	1.10
(3.41, 3.68]	3.52	26.9%	67.3%	1.21
(3.68, 4.00]	3.93	25.5%	65.3%	1.25

Table A3. Per-task-category triage lift under a global CP-width threshold (R2)

The selective-prediction analysis stratified by the paper’s four-category taxonomy, using a single global CP-width threshold that yields roughly 8% pooled acceptance. Exact-accuracy lift (accepted subset vs all items) is positive in every category, and largest on Vision-Heavy tasks, the category with the widest average width: the judges are uncertain on most such items, but markedly more accurate on the tight-interval subset.

Task category	Exact-accuracy lift
Aesthetics / AI	$1.35\times$
General VQA	$1.32\times$
Knowledge / Web	$1.23\times$
Vision-Heavy	$1.45\times$

Table A4. RSG with item-level bootstrap CIs, and cross-judge stability (R4 / R5c)

Item-level bootstrap ($B = 2000$, items resampled within each judge–dataset cell) for the Pearson ρ , with the RSG interval propagated through the paper’s fixed per-dataset width from Table 3, $\text{RSG} = \rho - (1 - w/4)$. The excerpt below is for LLaVA-Critic; the full three-judge, 14-dataset (42-cell) table replaces Table 17 in the camera-ready. The implementation was checked against the Fisher- z analytic interval on representative cells (ChartQA $n=400$: bootstrap half-width 0.071 vs Fisher 0.073; WIT $n=399$: 0.094 vs 0.096; MathVista $n=789$: 0.063 vs 0.060).

Dataset	ρ (item-bs 95% CI)	Width (Table 3)	RSG (95% CI)
AesBench	+0.402 [+0.273, +0.513]	2.08	−0.078 [−0.207, +0.033]
WIT	+0.164 [+0.071, +0.259]	2.38	−0.241 [−0.334, −0.146]
ChartQA	+0.507 [+0.431, +0.574]	3.08	+0.277 [+0.201 , +0.344]
InfographicsVQA	+0.411 [+0.331, +0.489]	3.50	+0.286 [+0.206 , +0.364]
MathVista	+0.376 [+0.312, +0.437]	3.37	+0.219 [+0.155 , +0.280]

WIT’s RSG of −0.241 matches the paper’s Table 17 value of −0.242 (rounding). The three highest-RSG datasets (ChartQA, InfographicsVQA, MathVista) remain clearly positive even under these wider item-level intervals.

Cross-judge stability of RSG rankings. Spearman rank correlation between the 14-dataset RSG vectors of each judge pair. All three pairs are positively correlated, indicating the high-RSG datasets are largely judge-stable.

Judge pair	Spearman ρ (p)
LLaVA-Critic, Gemini	+0.84 ($p = 0.0002$)
Phi-4, Gemini	+0.73 ($p = 0.003$)
LLaVA-Critic, Phi-4	+0.51 ($p = 0.064$)

Table A5. Width normalization corrected to $\div(K - 1)$ (R3, WoGp)

Interval width as a percentage of the score range under the incorrect $\div K = \div 5$ convention (as printed in Table 7 and the abstract) and the correct $\div(K - 1) = \div 4$ convention for a 1 to 5 scale (width range $[0, 4]$). Values are derived from existing data with no re-runs.

Slice	Width	Paper $\div 5$ (wrong)	Corrected $\div 4$
MLLM-Judge (Table 7, R2CCP)	3.05	61%	76%
Polaris (Table 7, R2CCP)	0.68	14%	17%
AesBench (narrowest task)	2.08	42%	52%
InfographicsVQA (widest task)	3.50	70%	88%
ChartQA	3.08	62%	77%
MathVista	3.37	67%	84%

Under the correct $\div 4$ normalization the range across the 14 task categories is 52% (AesBench) to 88% (InfographicsVQA); the abstract’s “ $\sim 40\%$ / $\sim 70\%$ ” was unsupported under either convention.